A chiral 4-polytope with automorphism group Sz(32768)

Human Compression project

Abstract

We give explicit matrices for a chiral polytope of type $\{31, 31, 25\}$ whose automorphism group is $\mathrm{Sz}(2^{15})$. Its facet and vertex-figure groups are two conjugate subfield groups $\mathrm{Sz}(2^5)$, meeting in a split torus of order 31. The construction contradicts Theorem 1.1 in the specified preprint edition of Spiga's paper on Suzuki polytopes. All finite-field calculations are supplied in a short executable certificate. The proof of generation and of the intersection property uses the classical structure of Suzuki groups, not an enumeration of the ambient group.

Spiga's preprint [1], arXiv:2010.04080v1 of 8 October 2020, asserts that every chiral polytope whose automorphism group lies between a Suzuki simple group and its automorphism group has rank three. The question is attributed there to Hubard and Leemans [2]. The theorem below concerns that precise source edition. It does not assert that every later printing has the same text, or make a separate priority claim about the subsequent literature.

Theorem 1. *There exists a chiral abstract polytope of rank four and type* $\{31, 31, 25\}$ *with automorphism group* $\mathrm{Sz}(32768)$.

1 Coordinates and retained group structure

Let $k = \mathbb{F}_{2^{15}}$ and $\theta = 2^8$. We use the standard Suzuki matrices

$$U(x,y) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ x & 1 & 0 & 0 \\ x^{\theta+1} + y & x^{\theta} & 1 & 0 \\ x^{\theta+2} + xy + y^{\theta} & y & x & 1 \end{pmatrix}, \qquad H(t) = \mathrm{diag}(t^{\theta+1}, t, t^{-1}, t^{-\theta-1}),$$

where $x, y \in k$ and $t \in k^*$. Let J be the matrix with ones on the antidiagonal and zeroes elsewhere. These matrices generate $G = \mathrm{Sz}(k)$; see, for example, the coordinate description in Bray and Bäärnhielm [4]. Write $K = \mathbb{F}_{32}$ and $F = \mathrm{Sz}(K)$ for the standard subfield group. The restriction of the map $x \mapsto x^{\theta}$ to K is $x \mapsto x^8$, so the same formulas define F.

We use the following classical facts from Suzuki's subgroup classification [3]. The group $\mathrm{Sz}(q)$ is simple for $q \geq 8$, has order $q^2(q^2+1)(q-1)$, and acts doubly transitively on its ovoid. A nonidentity element of a split torus fixes exactly the two points defining that torus. The maximal subgroups are point stabilizers, the three types of odd-torus normalizers, and maximal Suzuki subfield groups. Their relevant orders are

$$q^2(q-1), \qquad 2(q-1), \qquad 4(q+\sqrt{2q}+1), \qquad 4(q-\sqrt{2q}+1).$$

A subfield group $\mathrm{Sz}(2^d)$ is maximal when the odd quotient of field exponents is prime. The centralizer of a nonidentity odd-order element is cyclic of odd order. Finally, $\mathrm{Out}(\mathrm{Sz}(2^N)) \cong C_N$ for odd N. These are borrowed structural inputs. None is inferred from the finite computation below.

The standard Bruhat decomposition, also retained, is

$$\mathrm{Sz}(K) = U(K)H(K) \;\sqcup\; U(K)H(K)JU(K). \tag{1}$$

The first cell is lower triangular. In the second cell the entry in row one, column four is nonzero. Both assertions follow immediately by multiplying the displayed matrices.

2 The explicit rotations

Take z to be the residue class of X in

$$k = \mathbb{F}_2[X]/(X^{15} + X + 1), \qquad \eta = z^{1057}. \tag{2}$$

The polynomial is irreducible, z has order 32767, and η has order 31. Thus $K = \mathbb{F}_2(\eta)$. Set

$$\lambda = z^{11} + z^7 + z^6 + z^4 + z^3 + z + 1. \tag{3}$$

For reference, its minimal polynomial over K is

$$T^3 + \eta^{12}T^2 + \eta^{11}T + \eta. \tag{4}$$

In particular $\lambda \notin K$. Define the involution

$$i(x, y) = U(x, y)^{-1} J U(0, 1) J U(x, y)$$

and put

$$\begin{aligned} &u = i(\eta^2, \eta^3), \quad v_0 = i(\eta^5, \eta^{30}), \\ &h = H(\lambda), \qquad v = h v_0 h^{-1}, \qquad b = H(\eta), \\ &a = ub^{-1}, \qquad\quad c = b^{-1}v. \end{aligned} \tag{5}$$

These formulas, together with (2)–(3), specify every entry without a choice of a primitive element or of an unspecified root.

Lemma 2. *The elements in (5) satisfy*

$$|a| = 31, \quad |b| = 31, \quad |c| = 25, \qquad |ab| = |bc| = |abc| = 2. \tag{6}$$

Neither u nor v_0 is upper triangular, lower triangular, or antidiagonal.

Proof. The assertions are exact calculations in the field (2). We give the calculation in executable form as `check_arithmetic.py`; it uses only polynomial arithmetic over $\mathbb{F}_2$ and multiplication of 4×4 matrices. An integer $m = \sum m_j 2^j$ denotes $\sum m_j z^j$. In this notation $\eta = 624$ and $\lambda = 2267$, and the two involutions occurring before conjugation are

$$u = \begin{pmatrix} 4671 & 4175 & 5401 & 1 \\ 5156 & 1562 & 1830 & 5401 \\ 845 & 5157 & 1562 & 4175 \\ 1131 & 845 & 5156 & 4671 \end{pmatrix}, \quad v_0 = \begin{pmatrix} 5401 & 5401 & 4867 & 1 \\ 4467 & 1563 & 1562 & 4867 \\ 1131 & 5717 & 1563 & 5401 \\ 624 & 1131 & 4467 & 5401 \end{pmatrix}. \tag{7}$$

Reduction modulo $X^{15} + X + 1$ gives $a^{31} = b^{31} = c^{25} = I$, with $a, b, c^5 \neq I$. It also gives $u^2 = v^2 = (abc)^2 = I$, with all three matrices nonidentity. The identities $ab = u$ and $bc = v$ follow from the definitions. The off-diagonal and diagonal entries in (7) establish the last assertion directly.

For reproducibility, the certificate first checks irreducibility by $z^{2^{15}} = z$ and the two polynomial gcd tests for the prime divisors $3, 5$ of 15. It verifies primitivity by the prime divisors $7, 31, 151$ of 32767, reconstructs every matrix from (2)–(5), and checks all the equalities just stated. It also verifies the Suzuki and symplectic matrix equations and rejects a deliberately altered entry of v. The earlier search output is not read by this checker. □

3 The two face groups and their intersection

Lemma 3. *We have* $\langle a, b\rangle = F$ *and* $\langle b, c\rangle = hFh^{-1}$.

Proof. The first group equals $\langle u, b\rangle \le F$. At $q = 32$, the two nonsplit torus normalizers have orders 100 and 164, and the proper Suzuki subfield group is Sz(2), of order 20. None has order divisible by 31. A proper subgroup containing b must therefore lie in a point stabilizer or in the normalizer of a split torus. The only ovoid points fixed by b are the standard points $0, \infty$. Their stabilizers are upper and lower triangular, respectively. The normalizer of $\langle b\rangle = H(K)$ consists of the diagonal matrices and their antidiagonal coset. Lemma 2 excludes every possibility for a proper subgroup containing u, b. This proves the first assertion.

The element h centralizes b. Consequently $h^{-1}\langle b, c\rangle h = \langle b, v_0\rangle$. The same argument, with v_0 in place of u, proves the second assertion. □

Lemma 4. *For the diagonal conjugator in (5),*

$$F \cap hFh^{-1} = H(K) = \langle b\rangle. \tag{8}$$

Proof. The diagonal subgroup $H(K)$ is contained in the intersection. Suppose $g \in F \cap hFh^{-1}$. Then both g and $h^{-1}gh$ have entries in K. If g belongs to the second Bruhat cell in (1), its $(1, 4)$ entry is in K^*, whereas conjugation by h multiplies this entry by $\lambda^{-2(\theta+1)}$. Hence $\lambda^{514} \in K$. Since $\gcd(514, 32767) = 1$, this implies $\lambda \in K$, a contradiction.

Thus $g = U(x, y)H(t)$ with $x, y \in K$ and $t \in K^*$. Direct conjugation gives

$$h^{-1}U(x, y)h = U(x\lambda^{\theta}, y\lambda^{\theta+2}). \tag{9}$$

The root parameters in (1) are unique, so the two parameters on the right belong to K. If $x \neq 0$, then $\lambda^{256} \in K$; if $x = 0$ and $y \neq 0$, then $\lambda^{258} \in K$. Both are impossible because $\gcd(256, 32767) = \gcd(258, 32767) = 1$. Therefore $x = y = 0$ and $g \in H(K)$, as required. □

Lemma 5. *The three rotations generate* $G = \mathrm{Sz}(32768)$.

Proof. They generate both face groups from Lemma 3. The first, $F = \mathrm{Sz}(32)$, is a maximal subgroup of G because the quotient of field exponents is $15/5 = 3$. Equation (8) implies that the second face group is distinct from F. Hence the generated subgroup properly contains F and must equal G. □

4 The polytope and its chirality

We recall the rank-four rotation criterion of Schulte and Weiss [5], in the explicit form recorded in Leemans, Moerenhout and O'Reilly-Regueiro [6, §2, (C3′)]. Given generators a, b, c of finite orders at least two, with ab, bc, abc involutions, the required intersection condition is

$$\langle a\rangle \cap \langle b\rangle = 1, \qquad \langle b\rangle \cap \langle c\rangle = 1, \qquad \langle a, b\rangle \cap \langle b, c\rangle = \langle b\rangle. \tag{10}$$

These conditions give a chiral or directly regular abstract polytope. It is chiral precisely when there is no involutory automorphism α of the generated group satisfying

$$\alpha(a) = a^{-1}, \qquad \alpha(b) = a^2 b, \qquad \alpha(c) = c. \tag{11}$$

This is a borrowed reconstruction theorem; the group and marking constructed in (5) are the new input.

Proof of Theorem 1. The rotation relations and generation follow from Lemmas 2 and 5. The groups $\langle a\rangle$ and $\langle b\rangle$ have prime order 31. If they had nontrivial intersection, they would coincide, and their product ab would have odd order, contrary to (6). Thus the first equality in (10) holds. The second follows from $\gcd(31, 25) = 1$, and the third is Lemma 4.

It remains to exclude (11). The outer automorphism group of G has odd order 15, so every involutory automorphism is inner. Write $\alpha(g) = s^{-1}gs$. Since G is centreless, $\alpha^2 = 1$ implies $s^2 = 1$. The automorphism cannot be the identity, because $|a| = 31$, so s would be an involution. But $\alpha(c) = c$ gives $s \in C_G(c)$, and this centralizer has odd order because $|c| = 25$. This is impossible. The polytope is therefore chiral, with full automorphism group G and type $\{31, 31, 25\}$. □

The computation also checks the two small dihedral groups $\langle a, bc\rangle$ and $\langle ab, c\rangle$, of orders 62 and 50, and their intersections with the face groups. Their mutual intersection has order two and is generated by abc. These finite checks corroborate (10); they do not replace the proof of (8) or the maximal-subgroup argument. The ambient group, of order 37 777 778 976 635 853 209 600, is never enumerated.

5 Comparison with the source coordinates

For a direct comparison with [1], let P be the permutation matrix for (1 2)(3 4). It preserves J and changes our torus to

$$PH(t)P^{-1} = \operatorname{diag}(t, t^{\theta+1}, t^{-\theta-1}, t^{-1}).$$

Our Suzuki graph map takes the minors with row and column pair order $(12, 13, 24, 34)$, then raises entries to 2^7. Conjugating by P changes that order to $(24, 12, 34, 13)$, exactly the order in Wilson's product as displayed on page one of [1]. This verifies the change of model rather than assuming identical coordinate conventions.

The certificate constructs $g \in G$ with $g^{-1}(abc)g = J$. In the source basis, the inverse of the first normalized rotation is

$$x = P(g^{-1}ag)^{-1}P^{-1} = \begin{pmatrix} 29124 & 17285 & 29707 & 20757 \\ 17285 & 3002 & 31396 & 7939 \\ 29707 & 31396 & 8273 & 19487 \\ 20757 & 7939 & 19487 & 23604 \end{pmatrix}. \tag{12}$$

The integer encoding is the one used in (7). This matrix is symmetric, belongs to the Suzuki group in the source model, and satisfies $xJx = J$. Nevertheless its entries give

$$x_{14}x_{24} + x_{12}x_{44} = 28331 \neq 0. \tag{13}$$

Thus it violates equation (13) of [1], whose right-hand side would be zero.

The reason is visible before that equation is imposed. The source states Wilson's condition $gx \bullet gy = g(x \bullet y)$ only for pairs with symplectic pairing zero. On page three it also imposes this condition on (e_1, e_4) and (e_2, e_3), both of which have pairing one. For the concrete matrix (12),

$$(xe_1) \bullet (xe_4) = (3245, 26415, 3338, 22277)^{\mathsf{T}} \neq 0, \qquad e_1 \bullet e_4 = 0.$$

The admissible orthogonal-pair identities do hold. The failure is therefore an application outside the stated domain of the product identity, not a discrepancy between the two matrix models.

There is also a count that detects the same defective classification. For $\mathcal{X} = \{x \in G : xJx = J\}$, the map $x \mapsto xJ$ is a bijection from $\mathcal{X}$ to the involutions together with the identity. Each of the q^2+1 ovoid points is fixed by $q-1$ involutions, and each involution has a unique fixed point. Hence $|\mathcal{X}| = (q^2 + 1)(q - 1) + 1$, whereas the classification in the source would give $|\mathcal{X}| = q$.

This observation diagnoses the source argument; Theorem 1 additionally verifies the polytope intersection and chirality conditions.

The contribution of this note is the explicit marking, the diagonal subfield intersection argument as applied to it, and the resulting counterexample to the stated rank bound. The matrix model, Suzuki subgroup classification and polytope reconstruction theorem are retained inputs with the attributions above. No whole-source acceptance, independent review, Lean verification, or compression measurement is claimed by this author packet itself.

References

[1] P. Spiga, *On the rank of Suzuki polytopes: an answer to Hubard and Leemans*, arXiv:2010.04080v1, 8 October 2020, Theorem 1.1.

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[4] J. N. Bray and H. Bäärnhielm, *A new method for recognising Suzuki groups*, arXiv:1706.05697v1 (2017), §1.

[5] E. Schulte and A. Ivić Weiss, *Chiral polytopes*, Applied Geometry and Discrete Mathematics, DIMACS Series 4 (1991), 493–516.

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